

```

/* CMP-135 SDTM DS Creation
Domain: DS*/;
options validvarname=upcase nofmterr dlcreatedir;

libname raw  "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
libname sdtm  "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";

libname valxpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ds.xpt";

libname sexpt xport
"/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt";

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```

%macro make_date(invar=, outdt=);
    &outdt = .;

    if vtype(&invar) = "N" then do;
        if not missing(&invar) then do;
            if abs(&invar) > 50000 then &outdt = datepart(&invar);
            else &outdt = &invar;
        end;
    end;
else do;
    _dtm = input(strip(&invar), anydtdtm.);
    if not missing(_dtm) then &outdt = datepart(_dtm);

    if missing(&outdt) then
        &outdt = input(strip(&invar), anydtdte.);
end;
%mend;

```

```

data se_prep;
    set sexpt.se;

    se_start_dt = input(substr(SESTDTC, 1, 10), yymmdd10.);
    se_end_dt   = input(substr(SEENDTC, 1, 10), yymmdd10.);

    format se_start_dt se_end_dt date9.;

    keep USUBJID EPOCH se_start_dt se_end_dt;
run;

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```

data ex_prep;
    set raw.ex;

    %make_date(invar=EXSTD TN, outdt=exdt);

    format exdt date9.;

    keep SUBJECT exdt;
run;

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proc sql;

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create table ex_first as
select
    SUBJECT,
    min(exdt) as rfstdt format=date9.
from ex_prep
where not missing(exdt)
group by SUBJECT;
quit;
```

```
data ds_ic;
set raw.enr;

length
    STUDYID   $6
    DOMAIN    $2
    USUBJID   $40
    DSTERM    $25
    DSDECOD   $25
    DSCAT     $18
    DSSCAT    $26
    EPOCH     $11
    DSSTDTC   $10;

STUDYID = "CMP135";
DOMAIN  = "DS";
USUBJID = catx("-", STUDYID, strip(SUBJECT));

DSTERM  = "INFORMED CONSENT OBTAINED";
DSDECOD = "INFORMED CONSENT OBTAINED";
DSCAT   = "PROTOCOL MILESTONE";
DSSCAT  = "INFORMED CONSENT";
EPOCH   = "";

%make_date(invar=CNSTDTN, outdt=DSSTDTC);

DSSTDTC = put(DSSTDTC, yymmdd10.);

sortcat = 2;

keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
     DSSTDTC DSSTDTC SUBJECT sortcat;
run;
```

```
data ds_eos;
set raw.ds;

length
    STUDYID   $6
    DOMAIN    $2
    USUBJID   $40
    DSTERM    $25
    DSDECOD   $25
    DSCAT     $18
    DSSCAT    $26
    EPOCH     $11
```

```

        DSSTDTC  $10;

STUDYID = "CMP135";
DOMAIN  = "DS";
USUBJID = catx("-", STUDYID, strip(SUBJECT));

DSTERM  = upcase(strip(DSDECOD));
DSDECOD = upcase(strip(DSDECOD));

if DSTERM = "COMPLETED STUDY" then DSTERM = "COMPLETED";
if DSDECOD = "COMPLETED STUDY" then DSDECOD = "COMPLETED";

DSCAT   = "DISPOSITION EVENT";
DSSCAT  = "END OF STUDY";

%make_date(invar=DSSTDAT, outdt=DSSTDTC);

DSSTDTC = put(DSSTDTC, yymmdd10.);

sortcat = 1;

keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
      DSSTDTC DSSTDTC SUBJECT sortcat;
run;



---


proc sort data=ds_eos nodupkey;
  by SUBJECT DSTERM DSDECOD DSSTDTC;
run;



---


data ds_dd;
  set raw.dsdd;
  if upcase(strip(DSDECOD)) in ("COMPLETED", "COMPLETED STUDY") then delete;
  length
    STUDYID  $6
    DOMAIN   $2
    USUBJID  $40
    DSTERM   $25
    DSDECOD  $25
    DSCAT    $18
    DSSCAT   $26
    EPOCH    $11
    DSSTDTC  $10;

STUDYID = "CMP135";
DOMAIN  = "DS";
USUBJID = catx("-", STUDYID, strip(SUBJECT));

DSTERM  = upcase(strip(DSDECOD));
DSDECOD = upcase(strip(DSDECOD));

DSCAT   = "DISPOSITION EVENT";
DSSCAT  = "STUDY DRUG DISCONTINUATION";

%make_date(invar=DSSTDTCN, outdt=DSSTDTC);

DSSTDTC = put(DSSTDTC, yymmdd10.);

```

```
sortcat = 1;

keep STUDYID DOMAIN USUBJID DSTERM DSDECOD DSCAT DSSCAT EPOCH
      DSSTDTC DSSTDT SUBJECT sortcat;

run;
```

```
data ds_all_pre;
  set ds_eos ds_dd ds_ic;
run;
```

```
proc sql;
  create table ds_with_rfstdt as
  select
    a.*,
    b.rfstdt
  from ds_all_pre as a
  left join ex_first as b
    on a.SUBJECT = b.SUBJECT;
quit;
```

```
data ds_with_dy;
  set ds_with_rfstdt;

  if not missing(DSSTDT) and not missing(rfstdt) then do;
    if DSSTDT >= rfstdt then DSSTDY = DSSTDT - rfstdt + 1;
    else DSSTDY = DSSTDT - rfstdt;
  end;
run;
```

```
proc sql;
  create table ds_with_epoch as
  select
    a.STUDYID,
    a.DOMAIN,
    a.USUBJID,
    a.DSTERM,
    a.DSDECOD,
    a.DSCAT,
    a.DSSCAT,
    case
      when a.DSCAT = "PROTOCOL MILESTONE" then ""
      else coalescec(b.EPOCH, "")
    end as EPOCH length=11,
    a.DSSTDTC,
    a.DSSTDY,
    a.DSSTDT,
    a.sortcat
  from ds_with_dy as a
  left join se_prep as b
    on a.USUBJID = b.USUBJID
  and a.DSSTDT >= b.se_start_dt
  and a.DSSTDT <= b.se_end_dt;
```

```
quit;
data ds_with_epoch;
  set ds_with_epoch;

  if USUBJID = "CMP135-0124-1511"
    and DSTERM = "COMPLETED"
    and EPOCH = "MAINTENANCE"
  then delete;

  if USUBJID = "CMP135-1087-1514"
    and DSTERM = "COMPLETED"
    and EPOCH = "TITRATION"
  then delete;
run;

proc sort data=ds_with_epoch out=ds_sorted;
  by STUDYID USUBJID sortcat DSSTDTC DSTERM;
run;

data sdtm.ds;
  retain
    STUDYID DOMAIN DSSEQ DSTERM DSDECOD DSCAT DSSCAT
    EPOCH DSSTDTC DSSTDY USUBJID;

  length
    STUDYID $6
    DOMAIN $2
    DSTERM $25
    DSDECOD $25
    DSCAT $18
    DSSCAT $26
    EPOCH $11
    DSSTDTC $10
    USUBJID $40;

  set ds_sorted;
  by STUDYID USUBJID;

  if first.USUBJID then DSSEQ = 1;
  else DSSEQ + 1;

  label
    STUDYID = "Study Identifier"
    DOMAIN = "Domain Abbreviation"
    DSSEQ = "Sequence Number"
    DSTERM = "Reported Term for the Disposition Event"
    DSDECOD = "Standardized Disposition Term"
    DSCAT = "Category for Disposition Event"
    DSSCAT = "Subcategory for Disposition Event"
    EPOCH = "Epoch"
    DSSTDTC = "Start Date/Time of Disposition Event"
    DSSTDY = "Study Day of Start of Disposition Event";
/*      USUBJID = "Unique Subject Identifier"; */

  keep
    STUDYID DOMAIN DSSEQ DSTERM DSDECOD DSCAT DSSCAT
    EPOCH DSSTDTC DSSTDY USUBJID;
```

```
run;

proc sort data=sdtm.ds out=ds_created;
  by STUDYID USUBJID DSSEQ;
run;

proc sort data=valxpt.ds out=ds_validate;
  by STUDYID USUBJID DSSEQ;
run;

ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DS_PROC_COMPARE.rtf";

title "PROC COMPARE: SDTM DS Created vs Validation DS";

proc compare base=ds_validate compare=ds_created listall criterion=0.0000001;
  id STUDYID USUBJID DSSEQ;
```